Industrial Internship Report on

1. Crop and Weed Detection using YOLOv8

2. Remaining Useful Life (RUL) Prediction using Machine Learning

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was (Tell about ur Project) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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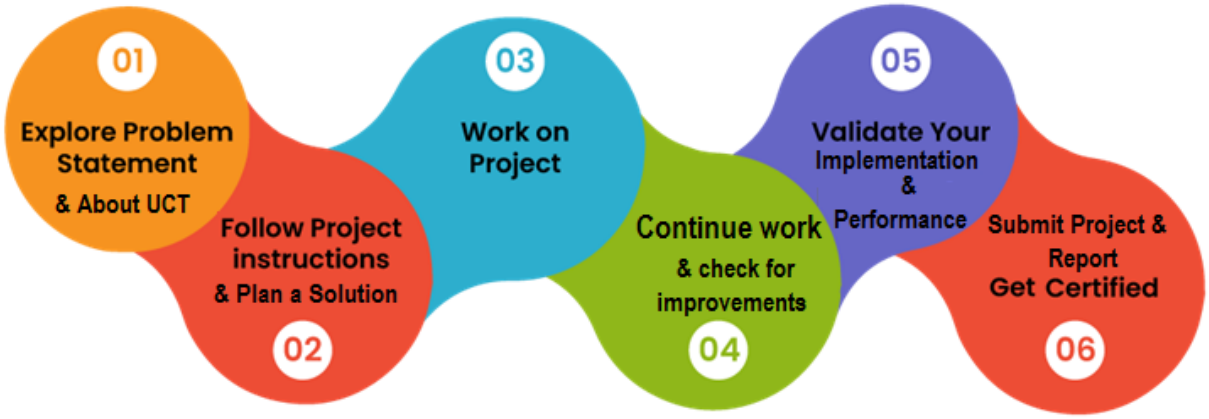
1 Preface

The four-week industrial internship conducted by Upskill Campus in collaboration with The IoT Academy and UniConverge Technologies Pvt Ltd (UCT) provided an excellent opportunity to gain practical exposure to real-world applications of machine learning and artificial intelligence. This internship was designed to bridge the gap between academic learning and industry requirements by focusing on hands-on project development.

During this internship, I worked on two significant projects: **Crop and Weed Detection using YOLOv8** and **Remaining Useful Life (RUL) Prediction using Machine Learning**. The first project focused on applying computer vision techniques to identify crops and weeds from agricultural images, aiming to support precision agriculture. The second project involved predicting the remaining useful life of industrial machinery using sensor data, which is crucial for predictive maintenance in industries.



Throughout this journey, I gained valuable insights into data preprocessing, model training, and performance evaluation. I also developed a deeper understanding of machine learning workflows and improved my analytical thinking.

I would like to express my sincere gratitude to Upskill Campus, The IoT Academy, and UniConverge Technologies Pvt Ltd for providing this opportunity. I am also thankful to my mentors and instructors for their guidance and support throughout the internship.

Finally, I would like to advise my juniors and peers to actively participate in such internships, as they provide practical exposure, enhance skills, and prepare students for real-world challenges.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

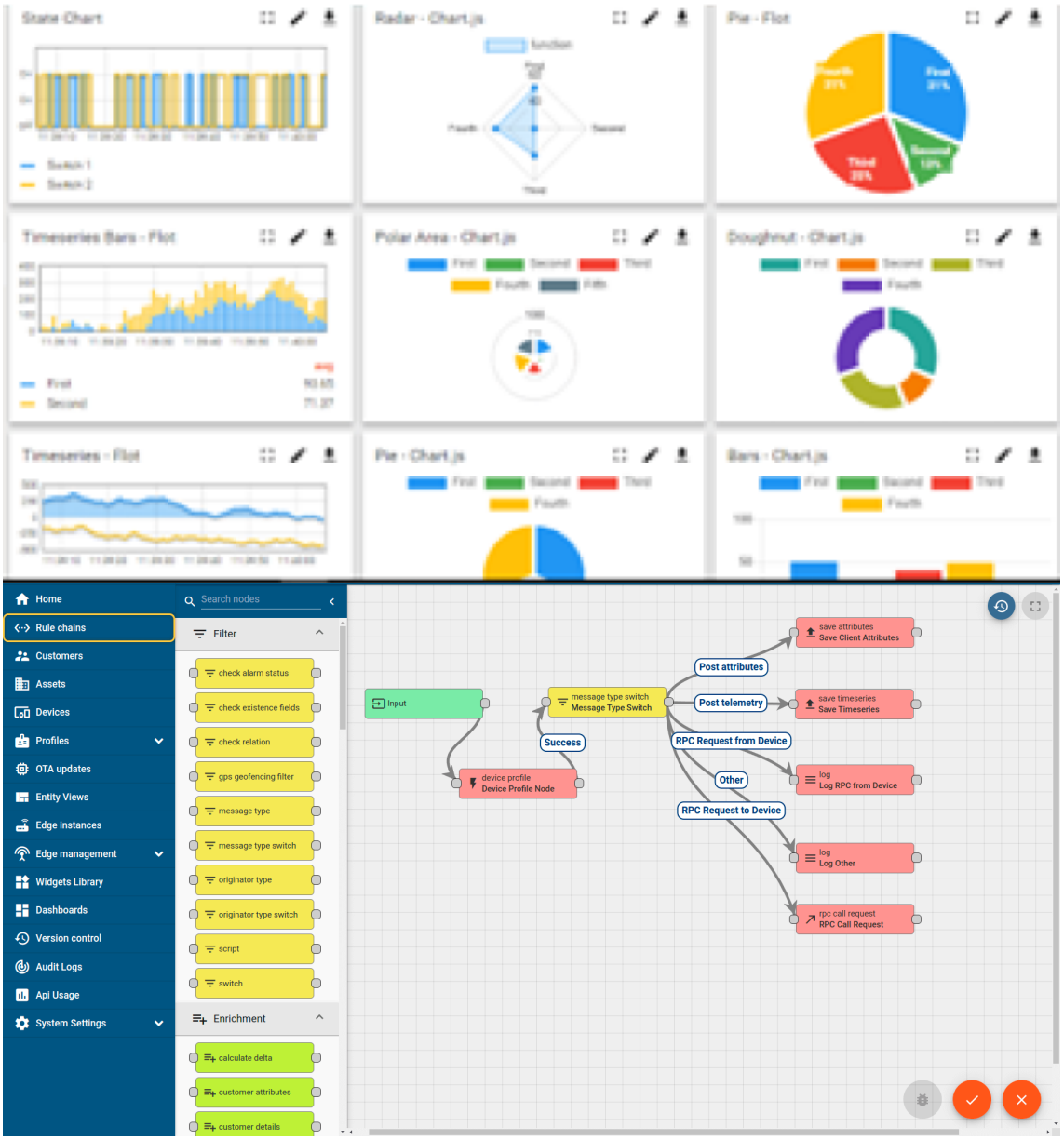
A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.
For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.
It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.

- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

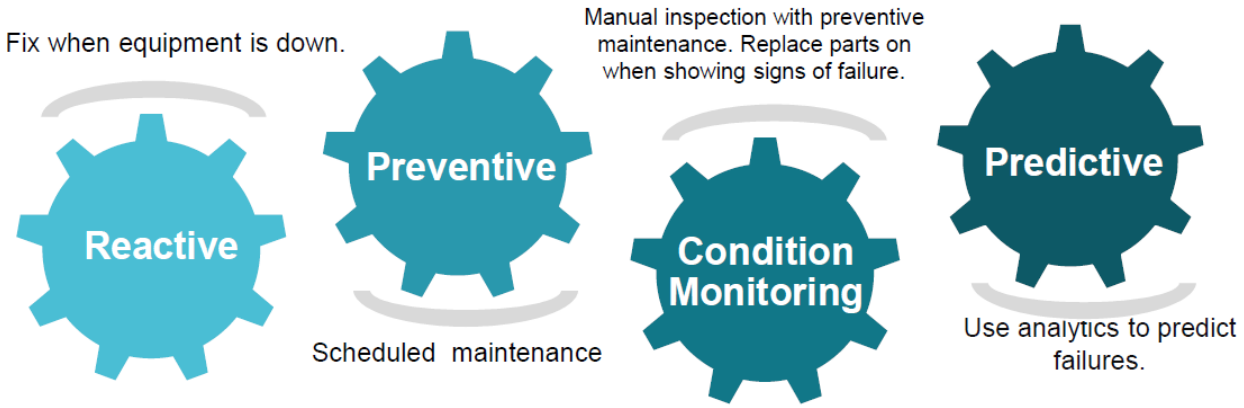
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

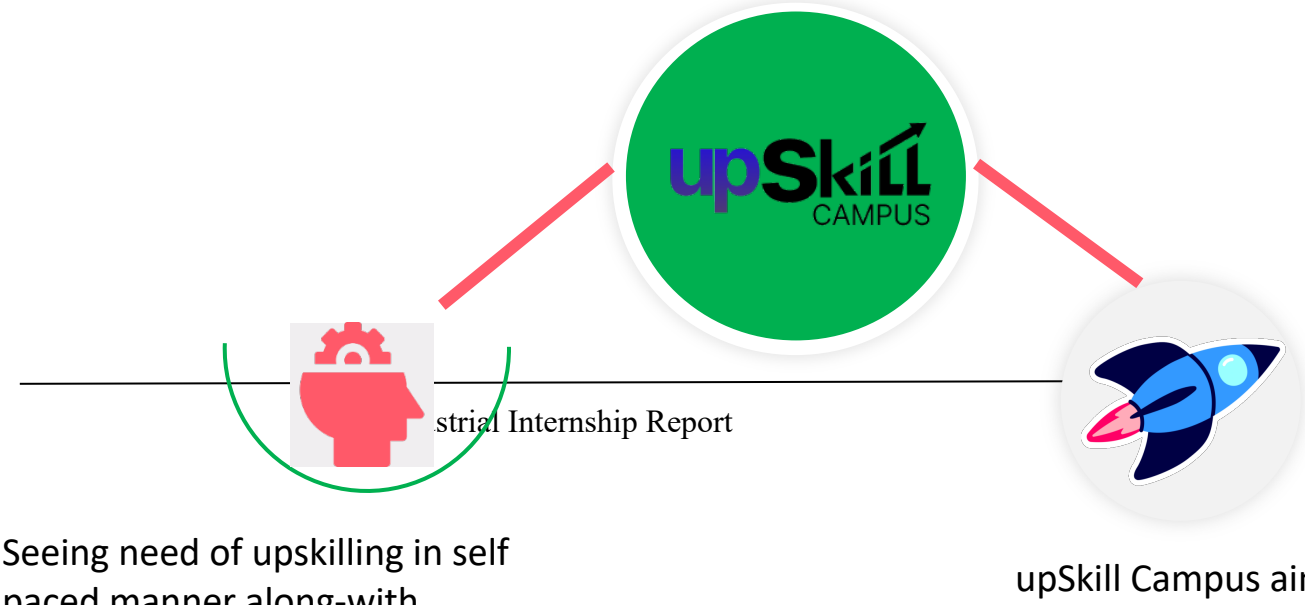
iv. Predictive Maintenance

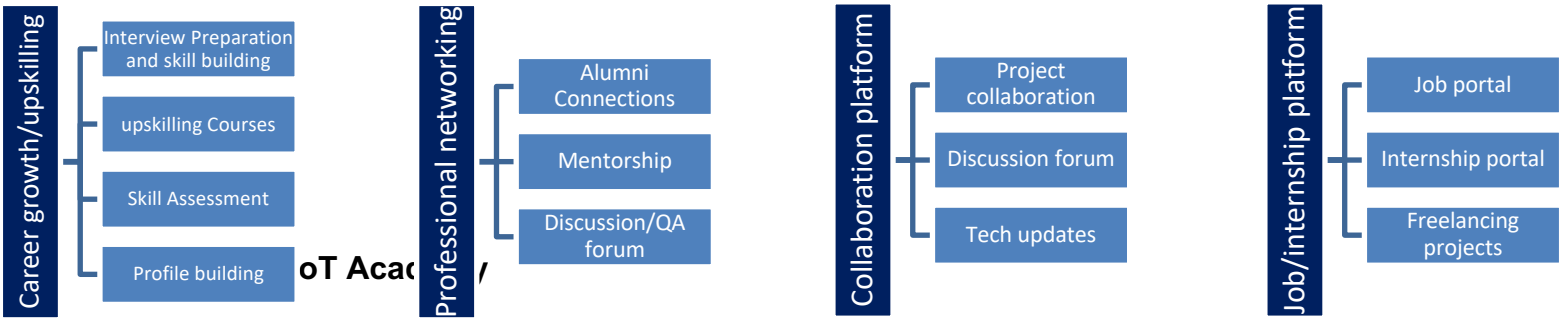
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process. USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.





The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Ultralytics. *YOLOv8 Documentation*.
Available at: https://docs.ultralytics.com
Description: This documentation was used to understand the YOLOv8 architecture, model training process, dataset formatting, and implementation details for object detection.

[2] Kaggle. *Crop and Weed Detection Data with Bounding Boxes*.
Available at: https://www.kaggle.com/datasets/ravirajsinh45/crop-and-weed-detection-data-with-bounding-boxes
Description: This dataset provided labeled images of crops and weeds in YOLO format, which were used for training and validating the object detection model.

[3] Scikit-learn Developers. *Scikit-learn: Machine Learning in Python*.
Available at: https://scikit-learn.org
Description: This library was used for implementing machine learning algorithms, data preprocessing, and evaluation metrics for the RUL prediction project.

2.6 Glossary

Terms	Acronym
Artificial Intelligence	AI
Machine Learning	ML
Deep Learning	DL
YouOnly Look Once (Object Detection Algorithm)	YOLO
Remaining Useful Life	RUL
Root Mean Square Error	RMSE
Convolutional Neural Network	CNN
Precision	PR
Recall	RC
Dataset	DS

3 Problem Statement

In the assigned problem statement, the objective was to solve real-world industrial problems using machine learning techniques.

The first problem focused on **Crop and Weed Detection** in agricultural fields. Identifying crops and weeds manually is time-consuming, labor-intensive, and prone to human error. There is a need for an automated system that can accurately detect and differentiate crops and weeds from images to support precision agriculture and improve productivity.

The second problem addressed **Remaining Useful Life (RUL) Prediction** of industrial machines. In industries, unexpected machine failures can lead to significant losses. Therefore, it is essential to predict the remaining useful life of machines using sensor data so that maintenance can be scheduled in advance, reducing downtime and improving efficiency.

Both problems required the development of intelligent systems capable of analyzing data and making accurate predictions. The aim was to design and implement machine learning models that can provide efficient, reliable, and scalable solutions to these challenges.

4 Existing and Proposed solution

Existing Solutions

In the domain of agriculture, traditional methods for crop and weed identification rely heavily on manual inspection. Farmers visually inspect fields, which is time-consuming, labor-intensive, and prone to human error. Some existing image processing techniques use basic filtering and thresholding methods, but they lack accuracy and robustness under varying environmental conditions.

In the case of industrial maintenance, conventional approaches follow reactive or preventive maintenance strategies.. Preventive maintenance, although better, is based on fixed schedules and does not consider the actual condition of the machine, resulting in inefficient resource utilization.

Limitations of Existing Solutions

- High dependency on manual effort
- Low accuracy in complex environments
- Lack of real-time prediction capabilities
- Inefficient maintenance scheduling in industries

Proposed Solution

To overcome these limitations, the proposed solution involves implementing machine learning-based approaches:

- For **Crop and Weed Detection**, the **YOLOv8 object detection model** is used to automatically detect and classify crops and weeds from images with high accuracy and speed.
- For **RUL Prediction**, a **machine learning regression model** is used to analyze sensor data and predict the remaining useful life of machinery.

4.1 Code submission:

https://github.com/sumantkg/upskillcampus/blob/main/Crop_and_Weed_Detection.py

https://github.com/sumantkg/upskillcampus/blob/main/Turbofan_Engine_RUL_Prediction.py

4.2 Report submission :

https://github.com/sumantkg/upskillcampus/blob/main/CropandWeedDetection_RULPrediction_Sumant_USC_UCT.pdf

5 Proposed Design/ Model

The proposed solution consists of two machine learning-based systems designed to solve real-world problems in agriculture and industrial maintenance. The overall design follows a standard machine learning pipeline, including data collection, preprocessing, model training, validation, and result generation.

For the **Crop and Weed Detection system**, images are used as input, and the YOLOv8 object detection model processes these images to identify and classify crops and weeds by drawing bounding boxes around detected objects.

For the **RUL Prediction system**, sensor data from machines is used as input. This data is preprocessed and fed into a machine learning regression model to predict the remaining useful life of the machine.

The workflow ensures efficient data handling, accurate model training, and reliable output generation.

5.1 High Level Diagram (if applicable)

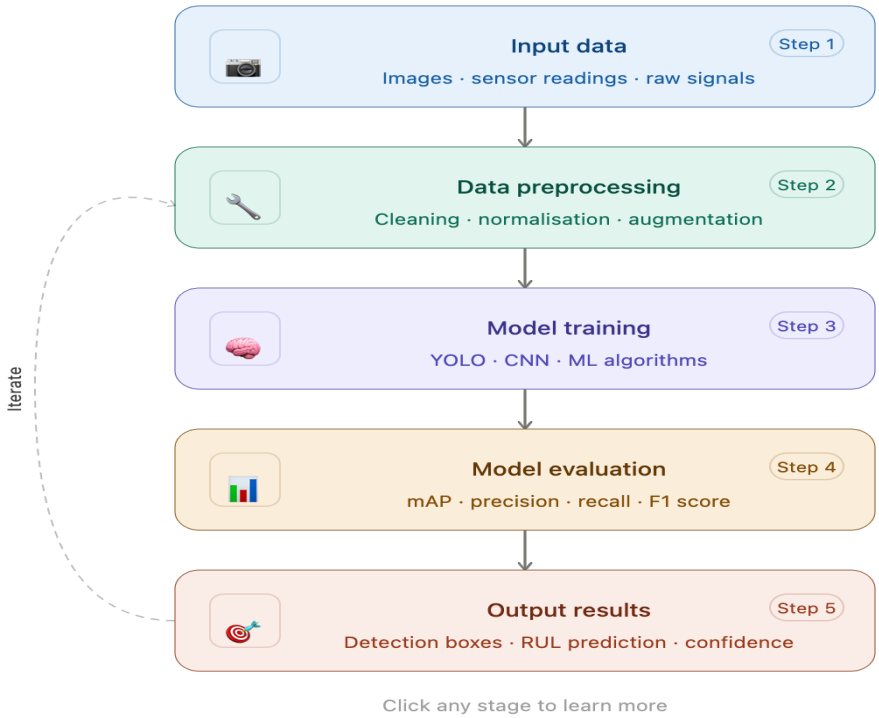


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

6 Performance Test

Performance testing ensures the models are effective in real-world industrial settings — not just theoretically sound.

Constraints & how they were handled

Constraint	Approach
Accuracy	YOLOv8 for detection, Random Forest for RUL prediction
Compute (CPU-only)	Lightweight YOLOv8n + optimized image/batch sizes
Memory	Efficient data handling and preprocessing
Scalability	Preprocessing pipeline designed for larger datasets
No GPU	Random Forest chosen for low computational cost

6.1 Test Plan/ Test Cases

Crop & Weed Detection:

- Tested on multiple images
- Verified bounding box accuracy

RUL Prediction:

- Trained on historical data
- Tested on unseen data and compared results

6.2 Test Procedure

- Split data into training and validation sets
- Trained models with selected parameters
- Evaluated results using visual outputs and metrics

6.3 Performance Outcome

Crop & Weed Detection:

- Accurate detection with bounding boxes
- Training time: ~20–30 minutes (CPU)

RUL Prediction:

- Effective RUL prediction
- Evaluated using RMSE with close predictions

7 My learnings

During the course of this internship, I gained valuable technical and practical knowledge that significantly enhanced my understanding of machine learning and its real-world applications.

I developed a clear understanding of **object detection using YOLOv8**, including how to work with labeled datasets, train models, and evaluate their performance. I also learned how to handle dataset preprocessing, manage directory structures, and troubleshoot common issues during implementation.

Through the **RUL Prediction project**, I gained experience in working with sensor data, applying machine learning algorithms, and evaluating model performance using metrics such as RMSE. This helped me understand how predictive analytics can be used in industrial maintenance to reduce downtime and improve efficiency.

Additionally, I improved my skills in:

- Data preprocessing and cleaning
- Model training and evaluation
- Debugging and problem-solving
- Working with Python libraries such as NumPy, Pandas, Scikit-learn, and YOLO

This internship also helped me understand the complete workflow of a machine learning project, from problem definition to final implementation. It improved my analytical thinking, patience, and ability to work on real-world challenges.

Overall, this experience has strengthened my technical foundation and prepared me for future opportunities in the field of machine learning and artificial intelligence.

8 Future work scope

The current implementation of the projects provides a strong foundation, but there are several opportunities for further improvement and enhancement.

For the **Crop and Weed Detection system**, future work can focus on using a larger and more diverse dataset to improve model accuracy and generalization. Training the model on GPU-based systems and increasing the number of epochs can further enhance performance. The system can also be extended for real-time deployment using drones or IoT-based devices for smart agriculture applications.

For the **RUL Prediction system**, more advanced machine learning or deep learning models such as LSTM can be implemented to capture time-series patterns more effectively. Incorporating real-time sensor data and deploying the model in industrial environments can improve its practical usability. Additionally, improving feature engineering techniques can lead to better prediction accuracy.

Further improvements for both projects include optimizing performance, reducing computation time, and integrating the solutions into scalable applications. These enhancements can make the systems more robust and suitable for real-world industrial deployment.